

Question	Answer	Marks
4(a)	pulse (of ultrasound)	B1
	* produced by quartz crystal/piezo-electric crystal	
	* gel/coupling medium (on skin) used to reduce reflection at skin	
	reflected from boundaries (between media)	B1
	reflected pulse/wave detected by (ultrasound) transmitter	B1
	reflected wave processed and displayed	B1
	* intensity of reflected pulse/wave gives information about boundary	
	* time delay gives information about depth of boundary	
	<i>max. 2 of additional detail points marked *</i>	B2
4(b)	$I_T = I_0 \exp(-\mu x)$	C1
	$2.9 = \exp(4.6\mu)$	C1
	$\mu = 0.23 \text{ cm}^{-1}$	A1

9702/42

Cambridge International AS/A Level – Mark Scheme

May/June 2017

PUBLISHED

Question	Answer	Marks
5(c)	any two reasonable suggestions e.g.	20